

Introduction to Cognitive Neuroscience

Psychophysics

Spring 2025

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Abstract

In this assignment, we dive into the science of how our brains decode facial identity under varying spatial frequencies! Fit psychometric curves, measure ROC performance, and propose your own cutting-edge hypothesis.

Facial recognition plays a crucial role in social cognition, with our brains having an extraordinary ability to process and interpret various facial features, such as identity, expression, and emotion. However, the underlying mechanisms of this process remain largely unexplored. In this assignment, we will examine the impact of different spatial frequencies on facial identity recognition. By applying filters to images of various individuals, we will create new versions of these images that selectively remove certain spectral information, resulting in images that are either smoother or less detailed than the original. Using these filtered images, we will explore how removing specific spatial frequencies influences our ability to recognize facial identity. This assignment will provide a deeper insight into the intricate processes involved in facial recognition and the specific role that spatial frequencies play in this cognitive task.

Task Description

To investigate the effects of spatial frequency on identity detection, we designed a psychophysics task featuring two pairs of male and female faces. Each pair was morphed across seven levels, with the first and last levels representing the original images. This resulted in 14 images at different morphing levels, which were then processed with low and high spatial frequency filters, creating a total of 42 stimuli. These stimuli were divided into three spatial frequency conditions (intact, high, and low) for the task. Participants were first given time to familiarize themselves with the names associated with each face, followed by a training phase using extreme morph levels to ensure proper recognition. The task consists of three blocks, each focused on a specific spatial frequency, and an additional fourth block combining all frequency groups. The order of blocks and hands used in the task is randomized for each participant.

The task consists of 1344 trials (42 images x 32 repetitions) spread across four blocks. In each trial, one of the stimuli is displayed at the center of the screen for 400 milliseconds, followed by two name options (male or female) on the left and right sides. Participants have three seconds to choose the name they believe matches the displayed face. The blocks are gender-controlled, ensuring that the options are always from the same gender. The task also includes a training phase where participants must correctly identify each face three times in a row before proceeding to the main task. The order of options within each block remains fixed.

Data Format

The data used in this assignment is stored in two CSV files:

1. **subjectInfo:** This file contains information about the participants. The "age" column refers to their age, the "Sex" column indicates their gender, and the "dom" column denotes their dominant hand.
2. **data:** This file reports the task performance. It contains details that differentiate each individual trial. The details of this file are as follows:
 - trialKeys: "MahGol" for trials in which the subject is choosing an image in the Mahnaz-Goli spectrum, and "AbHa" for trials in which the subject is choosing an image in the Abbas-Hasan spectrum.
 - levelFreq: spectral frequency level of presented stimulus (IF: intact, LF: low, HF: high)
 - levelFace: morph level, ranging from -3 to 3. Negative extremes show Abbas and Mahnaz, positive extremes show Ali and Goli for different pairs.
 - lCueName: the option showed to the subject on the left side of the screen.
 - rCueName: the option showed to the subject on the right side of the screen.
 - srespLoc: the key pressed by subject in this trial.
 - srespChoice: selected face ID.
 - RT: subject's reaction time in this trial.
 - Hand: the hand that subject used in this particular trial.
 - blockType: the type of block: The first three blocks of each subject are named "same" because all stimuli in these trials are from the same spectral frequency level. The fourth block is named "mix" because the trials are randomly selected from a pool of all images.
 - subjectID: a unique identifier for each subject.

Psychometric Fitting

In this section, we aim to fit a psychometric function to our data. This is one of the most important phases of analyzing cognitive task results, as the parameters obtained from the fit can assist us in testing various hypotheses and conducting statistical tests.

Here, we will plot the data based on the morph level and the target response choice. The morph level refers to the level of image morphing, which changes discretely from -3 to 3. The target response varies for each trial. For AbaHas, I choose Hasan as the target, and for MahGol, I select Goli. Since each image is repeated 32 times, we normalize the count of Hasan and Goli choices at each morph level by dividing by this number. This normalization process makes the data easier to work with in subsequent analyses. We will then modify the data according to the steps above and look at an example of a scatter plot for the first subject.

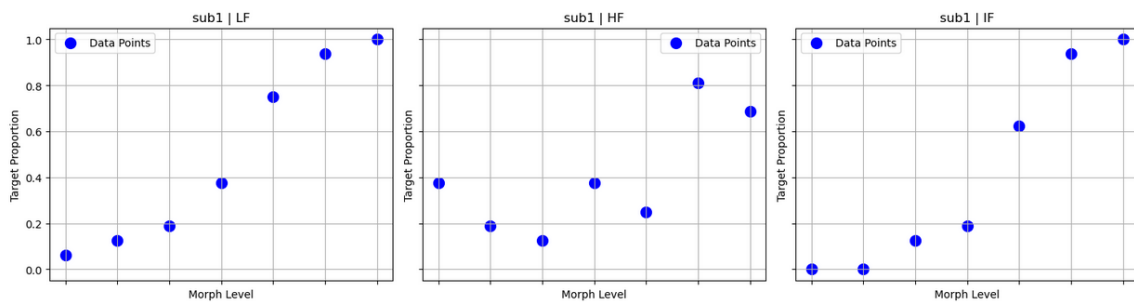


Figure 1 (Scatter Plot for the First Subject in Different Frequencies during the AbaHas Trial)

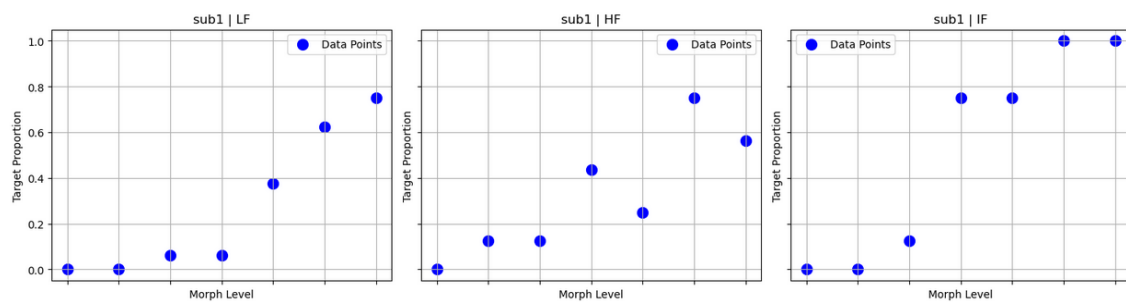


Figure 2 (Scatter Plot for the First Subject in Different Frequencies during the MahGol Trial)

Next, we proceed to fit the function to obtain parameters for further investigation. Based on the target scatter and the explanations provided in the project description, we can use the sigmoid function and similar functions. Using the scipy library, we use the following function with the parameters alpha and beta using the RSS (Residual Sum of Squares) algorithm.

$$\sigma(x) = \frac{\alpha}{1 + \exp(-\beta x)}$$

The results for the first subject are as follows:

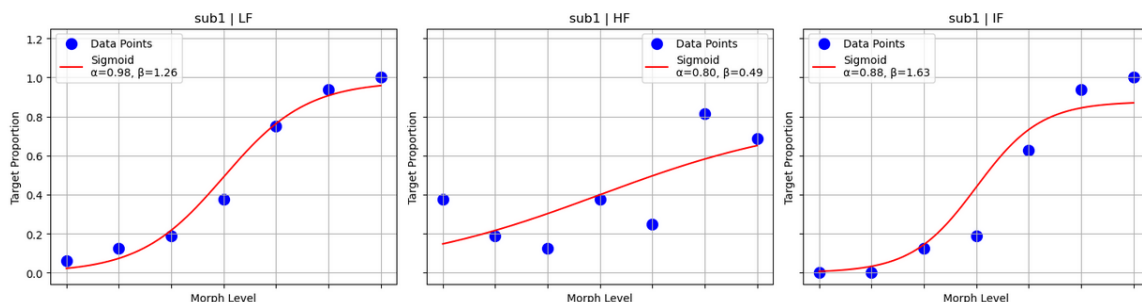


Figure 3 (Sigmoid Fitting for the First Subject in Different Frequencies during the AbaHas Trial)

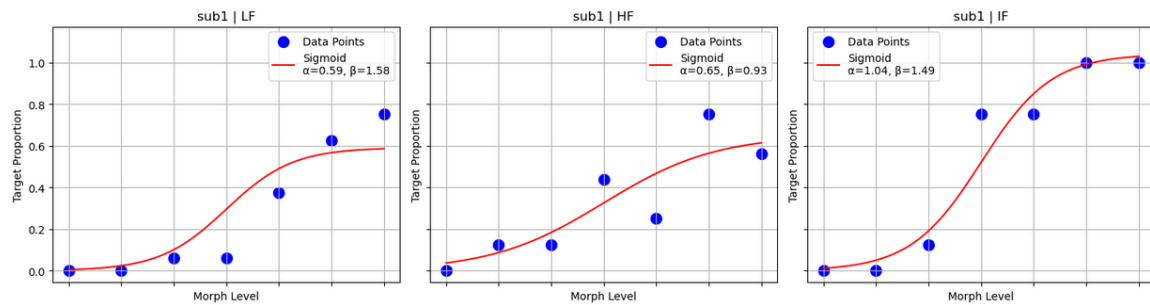


Figure 4 (Sigmoid Fitting for the First Subject in Different Frequencies during the MahGol Trial)

Additionally, we fit the following function in order to fit a better psychometric function. This function is a more general form of the sigmoid function. This is a modified sigmoid function. Due to having more parameters, it offers greater flexibility in fitting the data. However, it does not fit all datasets, and this potential error has been accounted for in the code.

$$\sigma(x) = \frac{\alpha}{1 + \exp(-\beta(x - y))} + \lambda$$

The results for the first subject are as follows:

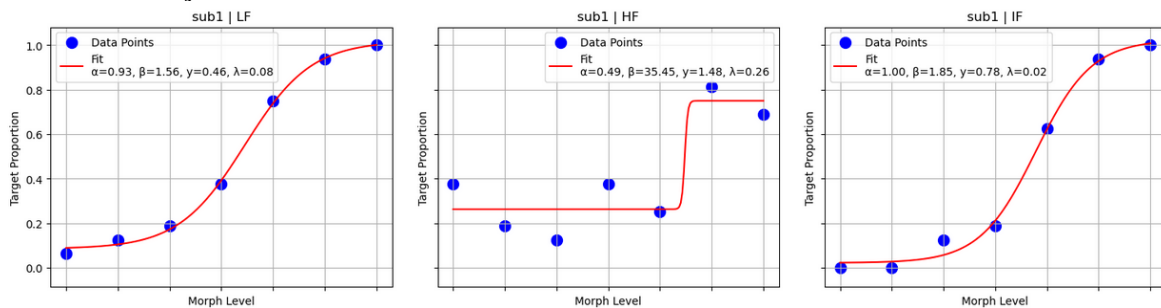


Figure 5 (Modified Sigmoid Fitting for the First Subject in Different Frequencies during the AbaHas Trial)

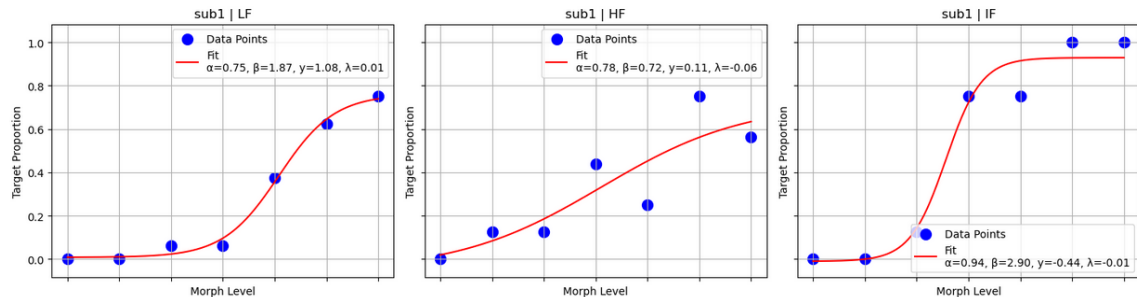


Figure 6 (Modified Sigmoid Fitting for the First Subject in Different Frequencies during the MahGol Trial)

Visually, it appears to fit the data better than the simple sigmoid function. However, for a more thorough evaluation, we use the Akaike Information Criterion (AIC) metric. AIC (Akaike Information Criterion) is a metric that balances goodness of fit and model complexity. It helps us select a model that describes the data well while using the fewest number of parameters possible. Compared to MSE, AIC offers important advantages because it takes into account not only the error of the model but also its complexity (i.e., the number of parameters). This feature allows AIC to strike a balance between fitting accuracy and model simplicity, thus preventing overfitting. Unlike MSE, which solely measures numerical error, AIC is based on the likelihood function and is more suitable for comparing models with different structures or complexities. For this reason, AIC is considered a more precise tool than MSE for model selection, especially in statistical and psychometric analyses.

Percentage of best model by AIC (AbaHas):

Modified_Sigmoid 64.04
sigmoid 35.96

Percentage of best model by AIC (MahGol):

Modified_Sigmoid 68.42
sigmoid 31.58

The results indicate that the modified model fits the data better than the simple sigmoid in most cases, though in some instances, the simple sigmoid performed more optimally.

In the next section, we will examine some hypotheses using the beta values obtained from the simple sigmoid function and perform the necessary statistical tests to infer whether the hypothesis is true or not.

First hypothesis:

Whether sensitivity to detecting identity from different spatial frequency bands differs.

First, we calculate the sensitivity (beta) for each subject. Note that the obtained beta values are for each trialKeys and frequencyLevel. Since we want to investigate the effect of sensitivity across different frequency levels (three frequency levels), we use the ANOVA test. ANOVA (Analysis of Variance) is used to compare the means of multiple groups. Since participants have been tested multiple times under different conditions, we use a Repeated Measures ANOVA.

The result for the trialKeys AbaHas is as follows:

```

Anova
=====
      F Value Num DF  Den DF Pr > F
-----
levelFreq   2.9686  2.0000 68.0000 0.0581
=====

```

The p_value is 0.0581, which indicates that the difference between frequencies in sensitivity to identity detection is close to significance, but still greater than the 0.05 threshold. Therefore, we cannot confidently say that there is a significant difference yet.

To compare each pairwise combination, we use a pairwise comparisons test. The result of this test is as follows.

```

LF vs HF: t=-1.6844, p=0.1013
LF vs IF: t=-1.8822, p=0.0684
HF vs IF: t=-1.5539, p=0.1295

```

As we can see, the p_values provided for each comparison are also above the threshold value. Therefore, this hypothesis does not hold for the trialKeys related to male faces.

Now, let's examine the trialKeys for MahGol. As before, after extracting the sensitivity, we first perform the ANOVA test. The result of this test is as follows:

```

Anova
=====
      F Value Num DF  Den DF Pr > F
-----
levelFreq   7.4617  2.0000 72.0000 0.0011
=====

```

Due to the p-value being smaller than 0.05, we proceed to investigate the different frequencies more closely, using the pairwise comparison test. This test shows which groups have significant differences. The result of this test is as follows:

```

LF vs HF: t=1.2370, p=0.2241
LF vs IF: t=-2.6767, p=0.0111
HF vs IF: t=-2.8039, p=0.0081

```

The result shows that the LP and HP frequencies do not differ significantly, but the sensitivity for these two is different from IF. To confirm this further, we use a more stringent statistical correction called Bonferroni. This test performs the pairwise test again with a new threshold. If the p_value is less than this new threshold, we can be more confident in the correctness of the hypothesis. The new threshold is obtained by dividing the original threshold (0.05) by the number of groups. The result of this test is as follows.

```

LF vs HF: t = 1.2370, p = 0.2241, corrected p = 0.6723, significant: False
LF vs IF: t = -2.6767, p = 0.0111, corrected p = 0.0334, significant: True
HF vs IF: t = -2.8039, p = 0.0081, corrected p = 0.0243, significant: True

```

Therefore, it can be concluded that sensitivity for the test related to the trialKey for females differs across different frequencies, and the stated hypothesis for female trialKeys is correct.

The reason we did not apply the Bonferroni correction for AbaHas is that the hypothesis related to it was already rejected, so there was no need to perform a more conservative test.

We can also use the box plot below to support the results of the statistical tests. For MahGol, there is a clear difference in the box plot of the IF frequency compared to the other frequencies, which visually confirms the results. However, for AbaHas, no obvious difference is observed in the plot.

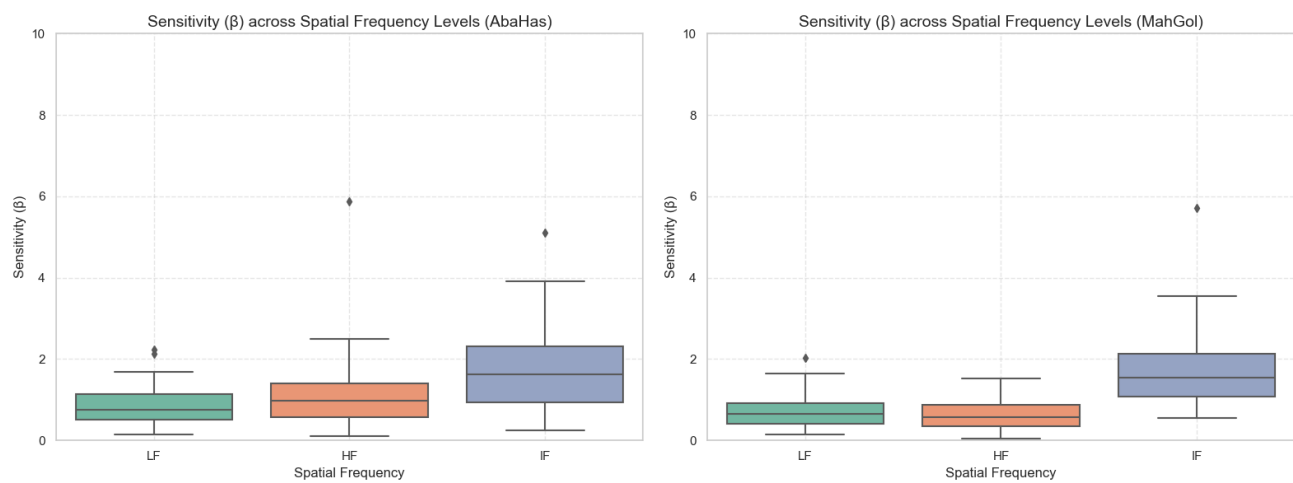


Figure 7 (Boxplot for AbaHas and MahGol Trials for Analyze First Hypothesis)

Second hypothesis:

Whether people are more adept at identifying images of their conspecific or hetero-specific gender.

Now we move on to testing the next hypothesis. This hypothesis investigates whether a person's gender influences their ability to recognize faces of their own gender. To explore this, we use data from the subjectInfo file. A new column called match is created: if the subject's gender matches the trial key, it is labeled Conspecific; otherwise, it is labeled Heterospecific.

To test this hypothesis, we use a t-test, as it is suitable for comparing sensitivity between two independent groups. The result of this test for AbaHas is as follows:

T-statistic: 0.712002647701969

P-value: 0.48001330645257945

The result shows that since the p-value = 0.618 is much greater than the conventional threshold of 0.05, we cannot conclude a significant difference in sensitivity between the Conspecific and Heterospecific groups.

The result for the MahGol trial key is as follows:

T-statistic: -1.0309288535601164

P-value: 0.3072293532757007

Since the p-value > 0.05, there is no significant difference in sensitivity between participants when recognizing faces of the same or opposite gender. This means that in the MahGol condition, the gender of the participant did not significantly affect their sensitivity to recognizing female faces.

We also plot boxplots to visually represent the results of the statistical tests.

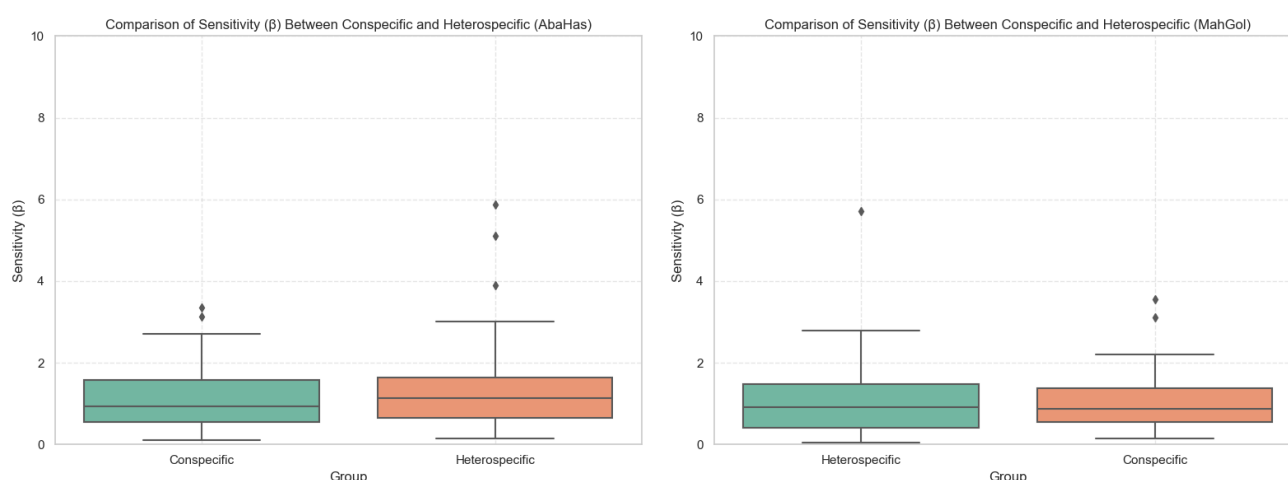


Figure 8 (Boxplot for AbaHas and MahGol Trials for Analyze Second Hypothesis)

Third hypothesis:

Whether women are significantly better than men in the task of identity detection.

This hypothesis examines identity recognition based on the participants' gender. Since we only have two groups of participants, we use a t-test to test the hypothesis. Based on the previous hypothesis, the gender data is complete. In this case, we combine both trials to make it different from the previous hypothesis. The test result is as follows:

T-statistic: -1.085801297966578

P-value: 0.2802413785224754

Based on the p-value, which is greater than 0.05, we reject the hypothesis. It should be noted that this hypothesis is generally examined for both men and women and is not dependent on the gender congruence of their responses. Therefore, gender does not have an effect on face recognition.

We also plot boxplots to visually represent the results of the statistical tests.

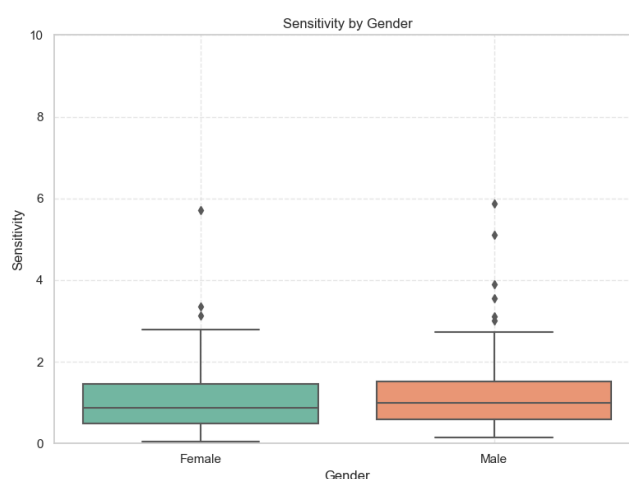


Figure 9 (Boxplot for Analyze Third Hypothesis)

I use the mean sensitivity across different levelFreqs and trialKeys, resulting in a dataset where each subject has a single sensitivity value. We then perform a t-test on this aggregated data. In this case, the variance used in the t-test changes, and the result differs slightly from before.

T-statistic: -1.1009601246092142

P-value: 0.2856762235204871

According to the p-value, this hypothesis is rejected with this different method of sensitivity calculation.

Fourth hypothesis:

Whether subjects can better detect identities in a specific spectral band if they use their left hand.

Before the test, we examine the hands in different ways but with the same idea for each subject and hypothesis. The idea is to extract the hand most frequently used to select the target from the initial dataframe (related to blocktype = same) and use that for analysis.

First, we examine the effect of the left and right hand in the trialkey for males. The results of the test are as follows:

Frequency	T-stat	P-value
LF	0.958604	0.345813
IF	-0.893188	0.383580
HF	0.342697	0.734882

The results of the t-test examining whether participants can better detect identities in a specific frequency band when using their left hands show that there is no significant difference in any of the frequency bands (LF, IF, HF), as all p-values are well above the 0.05 threshold. These findings suggest that using the left hand does not have a meaningful impact on identity sensitivity in any of the frequency bands in trialKey (AbHa). The box plot is shown below:

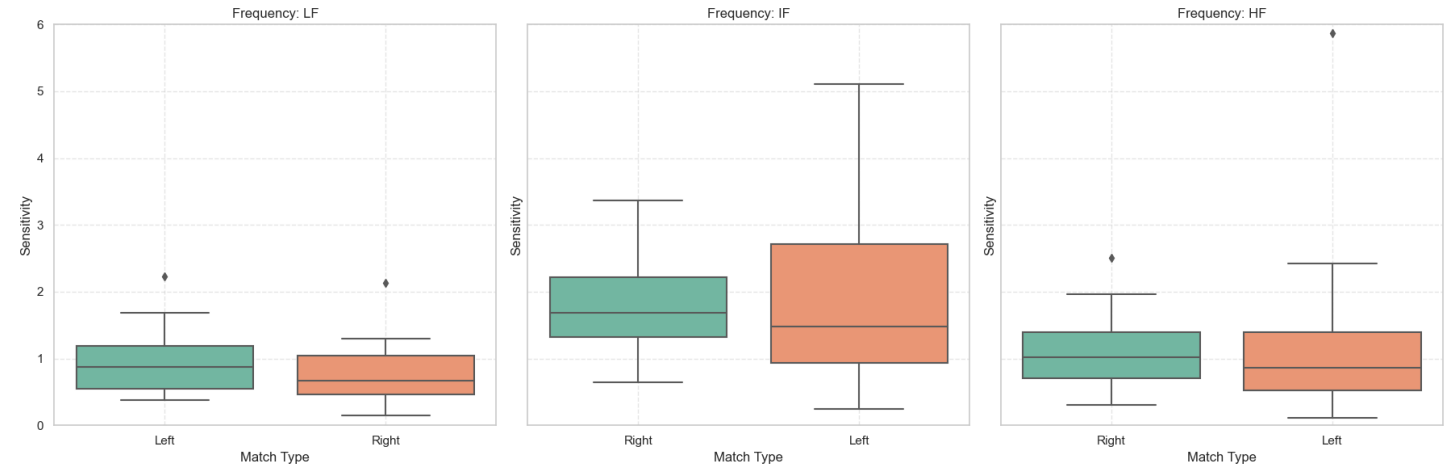


Figure 10 (Boxplot for Analyze Fourth Hypothesis)

The next step is to examine the effect of hand use in trialkey for females, and the result is as follows:

Frequency	T-stat	P-value
LF	-0.842770	0.405763
IF	-1.033596	0.315311
HF	-0.037400	0.970483

Once again, due to the similar reasoning, the hypothesis is rejected. The box plot is also shown below:

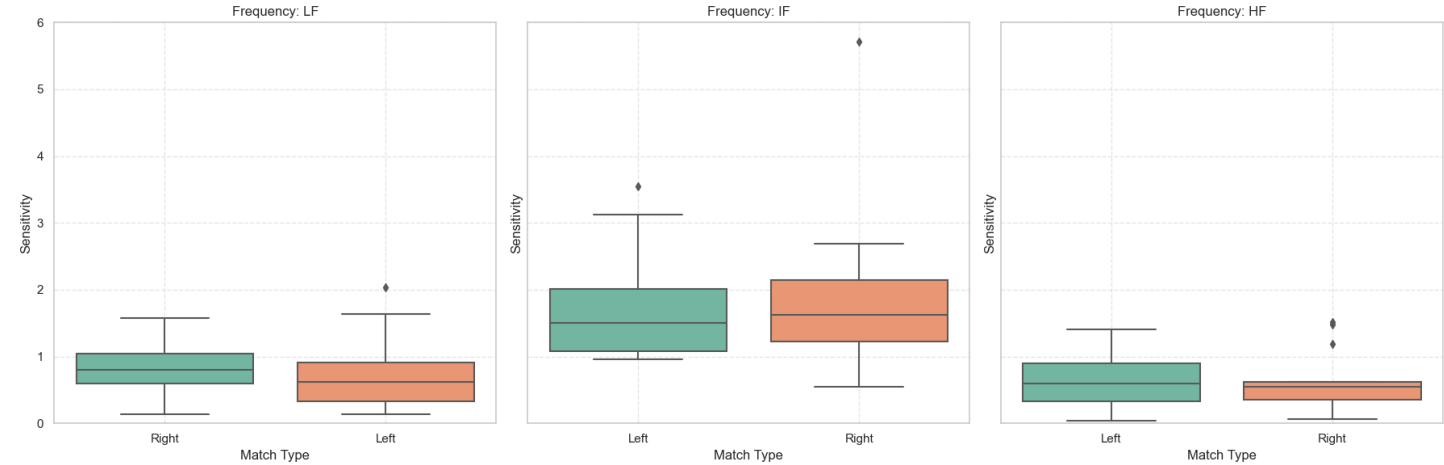


Figure 11 (Boxplot for Analyze Fourth Hypothesis)

We will test another hypothesis regarding the effect of hand on target selection (Hypothesis 10).

Fifth hypothesis:
Whether subjects can better detect identities in a specific spectral band if they use their dominant hand.

For this hypothesis, we again used the previous DataFrame and added the dominant hand information from subject_info to examine whether participants used their dominant hand. First, we assessed the effect of dominant hand usage in each frequency band for AbHa, with the results as follows:

Frequency	T-stat	P-value
LF	-0.945172	0.353686
IF	0.942427	0.357666
HF	-0.301940	0.765084

The t-test results for examining whether individuals perform better at identity recognition in specific frequency bands when using their dominant hand shows no significant differences in sensitivity across all three frequency levels (LF, IF, HF). The p-values are 0.35, 0.36, and 0.76 respectively, all of which are well above the conventional significance threshold of 0.05. Therefore, the hypothesis that using the dominant hand enhances recognition performance is not supported by the data in the AbHa condition. The corresponding boxplot is shown below:

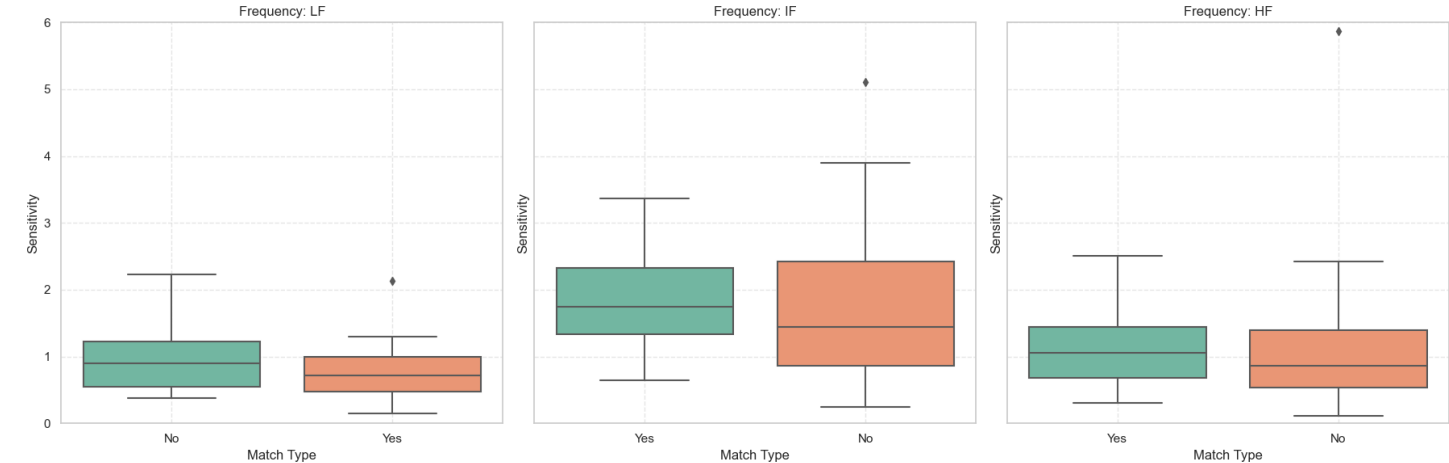


Figure 12 (Boxplot for Analyze Fifth Hypothesis (AbHa))

We conducted a similar analysis for MahGol:

Frequency	T-stat	P-value
LF	0.818133	0.420398
IF	1.084007	0.292085
HF	0.182009	0.856932

Again, no significant differences were observed in the MahGol condition. The boxplot is shown below:

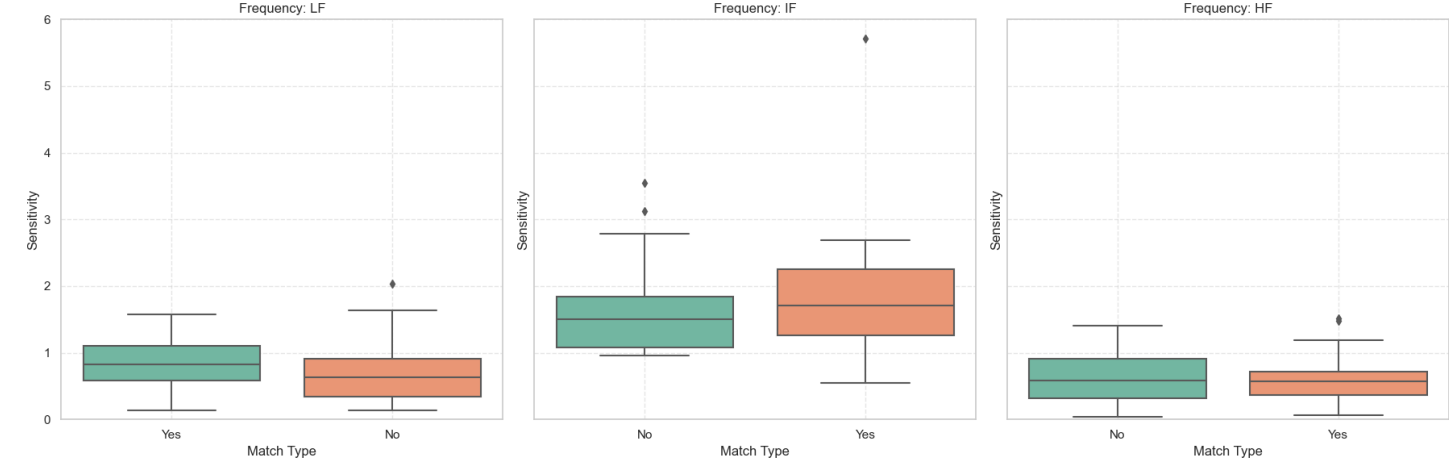


Figure 13 (Boxplot for Analyze Fifth Hypothesis (MahGol))

ROC Curve

The Receiver Operating Characteristic (ROC) curve is a tool used for evaluating the performance of models and decision-making systems. This curve provides a graphical representation of various rates of accuracy and error in an experiment or model. The horizontal axis of the curve represents the False Positive Rate (FPR), which is the ratio of incorrect positive responses to all samples that are truly negative. The vertical axis represents the True Positive Rate (TPR), which is the ratio of correct positive responses to all samples that are truly positive. The ROC curve is plotted based on changes in the model's decision threshold, showing the system's performance across different threshold levels.

The Area Under the Curve (AUC) of the ROC curve is a metric used to evaluate the performance of models. This value indicates the model's ability to correctly distinguish between different classes. AUC ranges from 0 to 1, with a value of 1 representing perfect separation between the two classes, and a value of 0.5 indicating random performance. The closer the AUC is to 1, the more sensitive and powerful the model is at distinguishing differences between the classes. This metric is particularly useful in situations where the class distribution is imbalanced.

The Area Under the ROC Curve (AUC) is recognized as a metric for the discriminatory power of models. In other words, AUC evaluates how well the model can differentiate positive samples from negative ones. Moreover, since AUC depends on the true and false positive rates rather than the exact number of predictions, it is useful for evaluating models in situations with class imbalance. Therefore, for classification and pattern recognition tasks, especially in fields such as medicine, engineering, and psychometrics, AUC serves as a key tool for comparing models and assessing their accuracy in detecting different patterns.

For each subject and each frequency level (levelFreq), an ROC curve is plotted to evaluate the individual's performance in identifying the target face. First, the data is filtered to include only the responses corresponding to that subject, frequency, and trial type (trialKeys). Then, a binary variable is created based on whether the response matches the target (Hasan or Goli). Using levelFace as the similarity score and the selected target labels, the ROC curve and AUC value are calculated and plotted. This curve demonstrates how well the subject can distinguish the target from non-targets under different conditions. The ROC curves for the first subject, separately for each trialKey, are shown below:



Figure 14 (ROC for the First Subject in Different Frequencies during the AbaHas Trial)

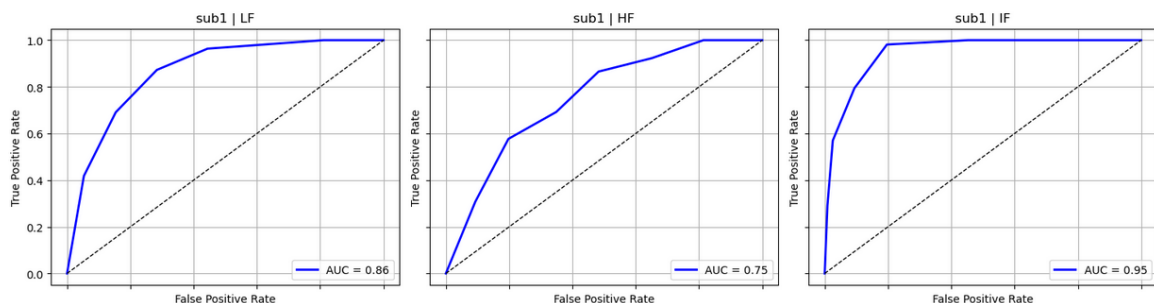


Figure 15 (ROC for the First Subject in Different Frequencies during the MahGol Trial)

This time, we use AUC as the measure of sensitivity to test the hypotheses. After evaluating the hypotheses, we will compare the results obtained using AUC with those obtained using sensitivity derived from the sigmoid fit.

First hypothesis:

Whether sensitivity to detecting identity from different spatial frequency bands differs.

Given the reasoning presented in the "first hypothesis" section, we again use repeated measures ANOVA when using AUC. The result of the analysis is as follows:

```

Anova
=====
      F Value Num DF  Den DF Pr > F
-----
levelFreq  1.4263  2.0000  74.0000 0.2467
=====

```

Based on the outcome, the p-value is very large, which doesn't support the hypothesis that frequency has an effect on AUC. We then proceed with a pairwise comparison test to determine which frequency pairs differ significantly in AbHa.

LF vs HF: $t=0.6971$, $p=0.4901$

LF vs IF: $t=-1.0596$, $p=0.2962$

HF vs IF: $t=-1.5472$, $p=0.1303$

The p-values for each pairwise comparison is also very large.

The boxplot illustrating this comparison is shown below:

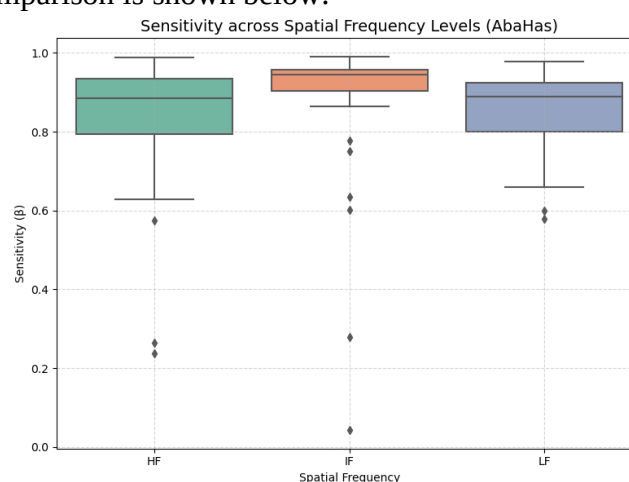


Figure 16 (Boxplot for Analyze First Hypothesis using AUC)

We also conduct the same analysis for the female trialKey, following the same reasoning as before. The result of the ANOVA is as follows:

```

Anova
=====
      F Value Num DF  Den DF Pr > F
-----
levelFreq 64.8757  2.0000  74.0000 0.0000
=====

```

This hypothesis is also supported due to the small p-value, and we can confidently state with 95% certainty that frequency has an effect on AUC. Again, we perform pairwise comparisons between frequencies to identify the differences among them.

LF vs HF: $t=0.7875$, $p=0.4360$

LF vs IF: $t=-11.0235$, $p=0.0000$

HF vs IF: $t=-11.1679$, $p=0.0000$

The p-values for each pairwise comparison is also very small. Specifically, the comparisons between LF/IF and HF/IF yield particularly low values, indicating a significant difference in AUC between those frequencies. However, the difference between LF and HF is less pronounced, suggesting a relatively weaker but still possible distinction in sensitivity for the female trialKey condition.

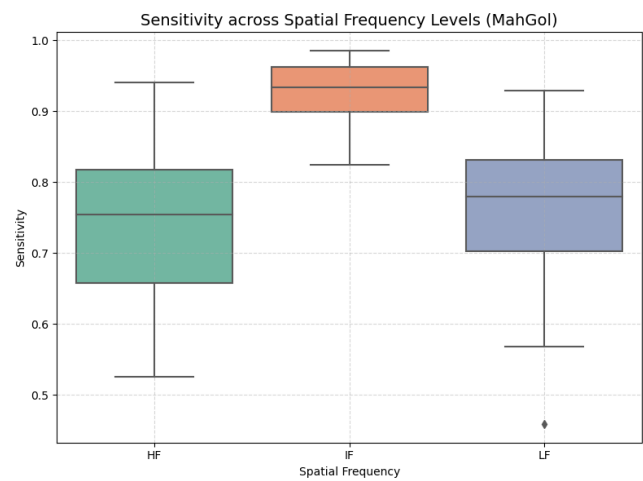


Figure 17 (Boxplot for Analyze First Hypothesis using AUC)

We also examine this hypothesis once more, this time without considering the trialKey. An important point to note is that we use the mean AUC for each subject at each levelFreq, since repeated measures ANOVA cannot handle having two AUC values per subject for each frequency level. The result of this test is as follows:

Anova					
	F	Value	Num DF	Den DF	Pr > F
levelFreq	25.2449	2.0000	74.0000	0.0000	

Once again, we can accept the hypothesis that there is a difference in AUC across different frequencies. The corresponding boxplot is shown below:

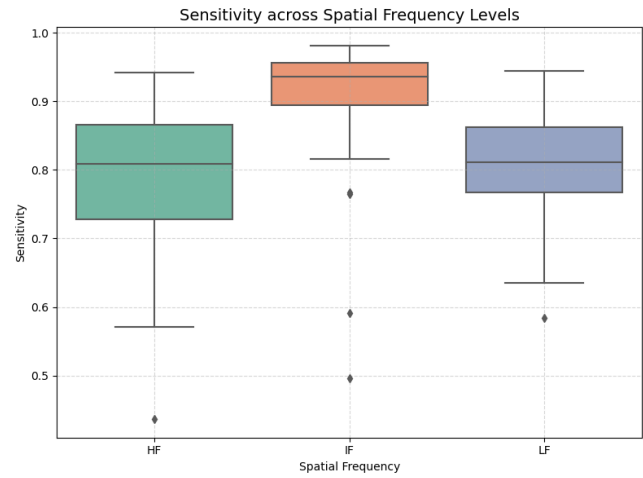


Figure 18 (Boxplot for Analyze First Hypothesis using AUC without consider trialKey)

Second hypothesis:

Whether people are more adept at identifying images of their conspecific or hetero-specific gender.

For this hypothesis, we first assigned labels indicating whether the subject and target were of the same gender in each trial. After this step, since we are comparing two independent groups, we used a t-test to evaluate the hypothesis. The result is as follows:

T-statistic: -1.915069868796347

P-value: 0.058956801200484264

Due to the p-value being more than 0.05, we can't confidently state that male participants performed better in recognizing individuals of their own gender in the AbaHas trialKey, and thus, the hypothesis isn't accepted. The corresponding boxplot is shown below:

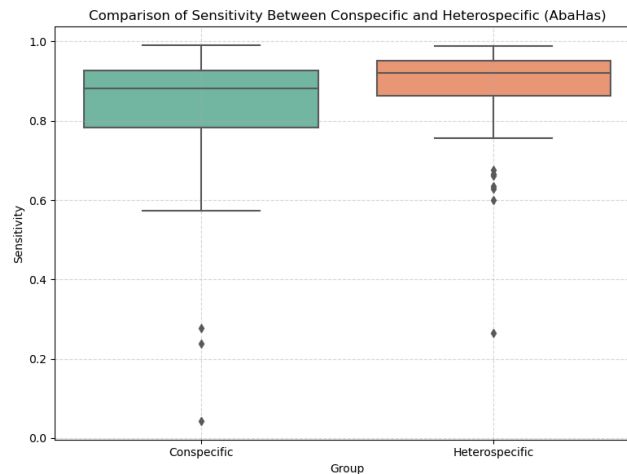


Figure 19 (Boxplot for Analyze Second Hypothesis using AUC for AbaHas)

We then performed the same analysis for the **female trialKey**. The result is as follows:

T-statistic: 1.2439127031684634

P-value: 0.21640261385752468

The p-value in this test is much larger than the threshold, meaning we can reject the hypothesis of same-gender recognition in this trialKey (MahGol). The corresponding boxplot is shown below.

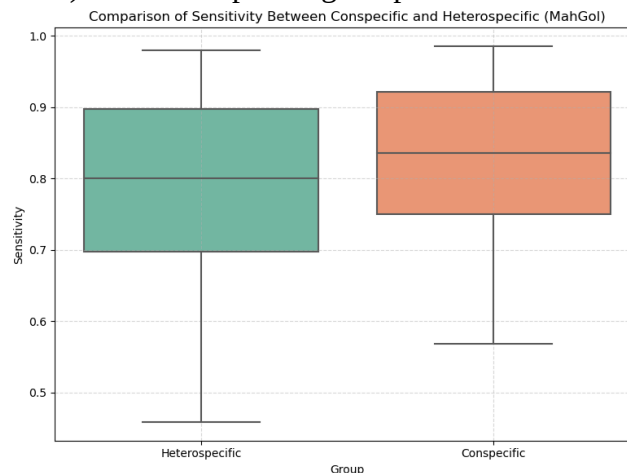


Figure 20 (Boxplot for Analyze Second Hypothesis using AUC for MahGol)

Next, we repeated the analysis **without considering the trialKey**. First, we calculated the average AUC across frequencies for each subject and used that as the subject's overall score. Note that this result reflects the general case, not a specific trialKey. The outcome is as follows:

T-statistic: -0.7791296023238015

P-value: 0.43870139839826106

Based on the resulting p-value, we reject the hypothesis that participants are better at recognizing individuals of the same gender across the whole task.

The corresponding boxplot is shown below.

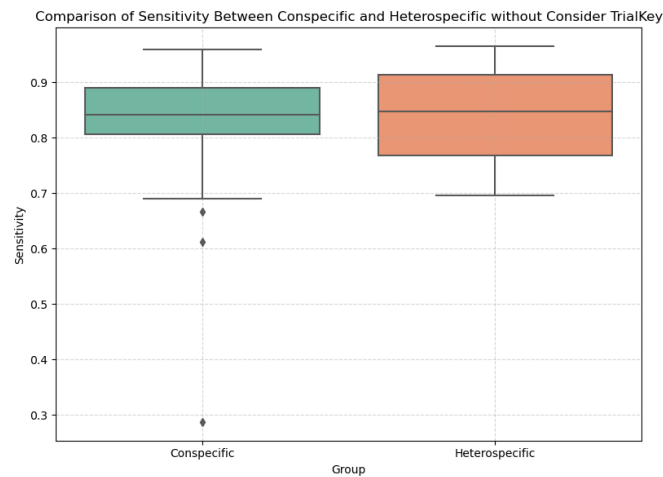


Figure 21 (Boxplot for Analyze Second Hypothesis using AUC without Consider TrialKey)

In the code, we also applied the averaging method for AUC values within each trialKey for the same-gender recognition hypothesis, and the test results also reject the hypothesis in those cases.

Third hypothesis:

Whether women are significantly better than men in the task of identity detection.

For this hypothesis, we use the same DataFrame as before. Here, we compare male and female participants in terms of their performance on the task, using their corresponding AUC values. Since we again have two independent groups, we use a t-test to test the hypothesis. The result is as follows:

T-statistic: 2.5682392565550276

P-value: 0.011046073143840686

The p-value indicates that women performed significantly better than men in this identity recognition task based on AUC. The corresponding boxplot is shown below:

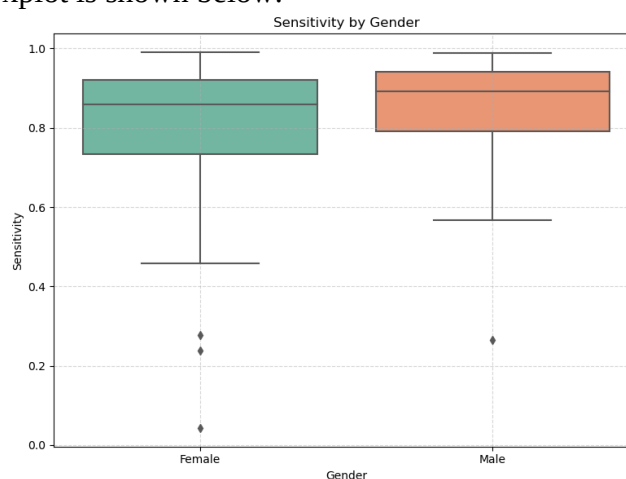


Figure 22 (Boxplot for Analyze Third Hypothesis using AUC)

For this hypothesis, I use the mean AUC across different levelFreqs and trialKeys, resulting in a dataset where each subject has a single AUC value. We then perform a t-test on this aggregated data. In this case, the variance used in the t-test changes, and the result differs slightly from before.

T-statistic: 1.9689208756739938

P-value: 0.05892780652906461

According to the p-value, this hypothesis is rejected with this different method of AUC calculation but it can accept with more trial.

Fourth hypothesis:
Whether subjects can better detect identities in a specific spectral band if they use their left hand.

First, we examine the effect of the left and right hand in the trialkey for males. The results of the test are as follows:

Frequency	T-stat	P-value
LF	0.958071	0.344968
IF	-0.401042	0.690816
HF	-0.754606	0.455503

The results of the t-test examining whether participants can better detect identities in a specific frequency band when using their left hands show that there is no significant difference in any of the frequency bands (LF, IF, HF), as all p-values are well above the 0.05 threshold. These findings suggest that using the left hand does not have a meaningful impact on identity sensitivity in any of the frequency bands in trialKey(AbHa). The box plot is shown below:

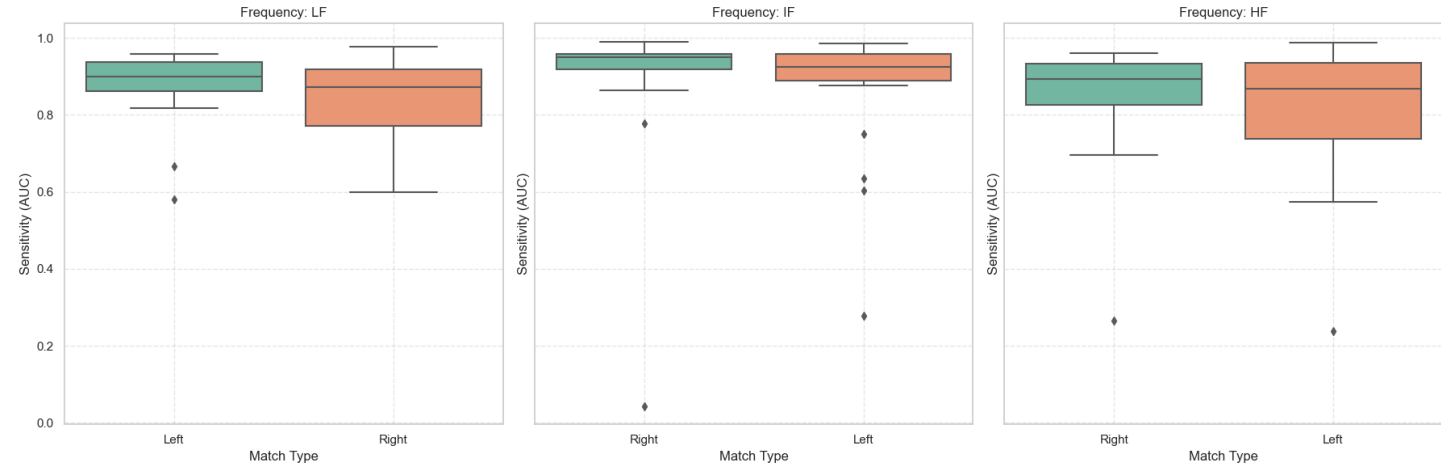


Figure 23 (Boxplot for Analyze Fourth Hypothesis)

The next step is to examine the effect of hand use in trialkey for females, and the result is as follows:

Frequency	T-stat	P-value
LF	-0.220683	0.827195
IF	-0.833653	0.410011
HF	-0.464204	0.646729

Once again, due to the similar reasoning, the hypothesis is rejected. The box plot is also shown below:

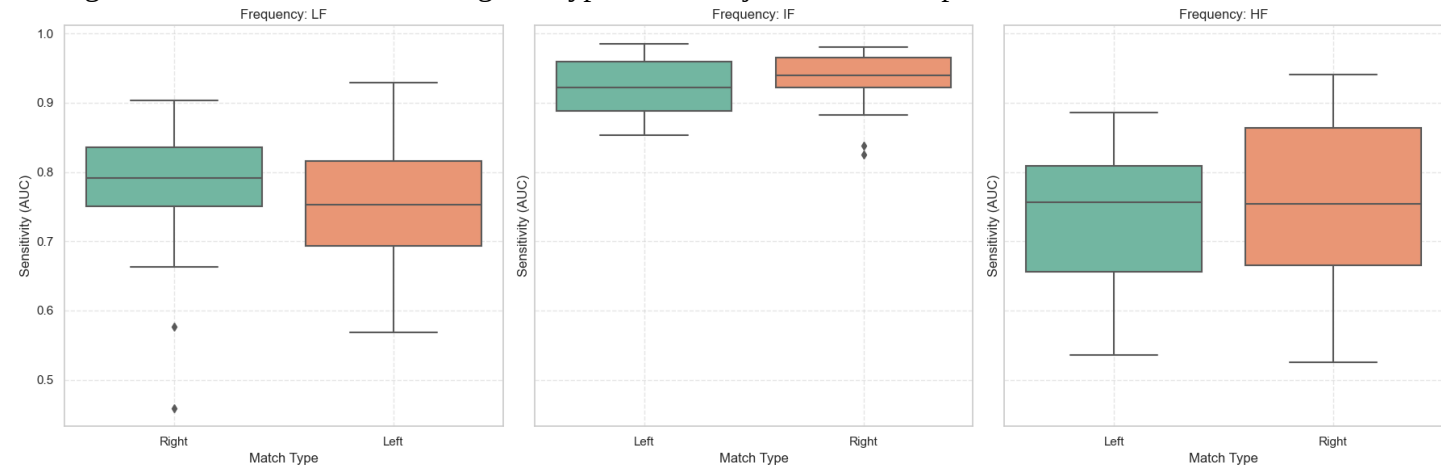


Figure 24 (Boxplot for Analyze Fourth Hypothesis)

We will test another hypothesis regarding the effect of hand on target selection (Hypothesis 10).

Fifth hypothesis:
Whether subjects can better detect identities in a specific spectral band if they use their dominant hand.

As with the sigmoid analysis, we follow the same approach here: we construct the relevant dataset and conduct the test to evaluate whether the hypothesis can be accepted or rejected.

Frequency	T-stat	P-value
LF	-0.759550	0.453458
IF	0.543242	0.590312
HF	0.584993	0.562201

As observed in AbHa, the use of the dominant hand does not result in any significant differences across the different frequency levels. However, a relatively greater difference is seen in the LF band compared to the other two frequency bands. The corresponding boxplot is shown below.

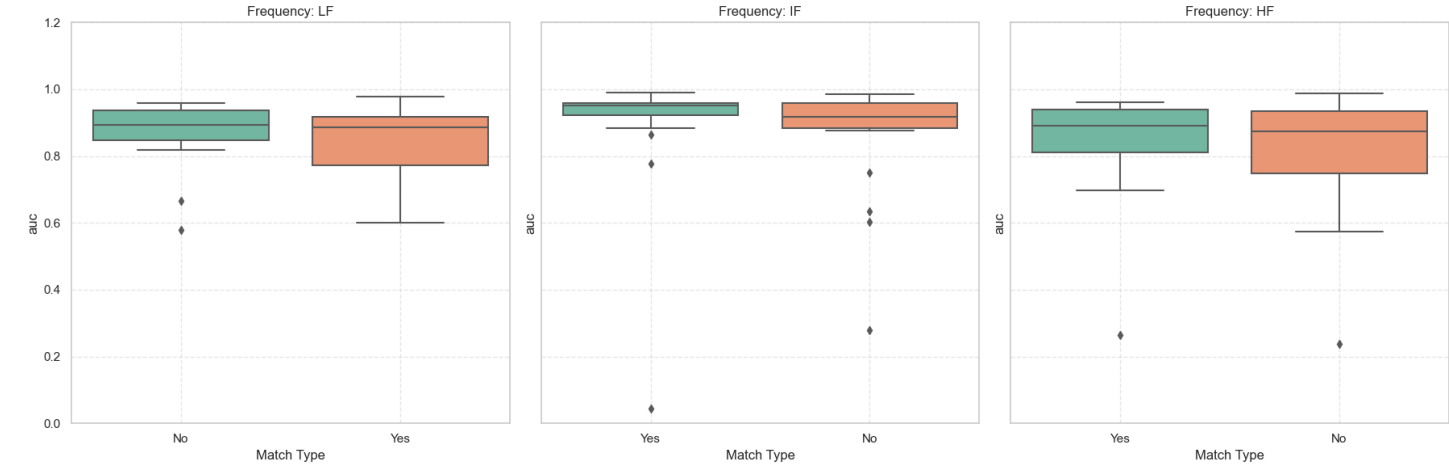


Figure 25 (Boxplot for Analyze Fifth Hypothesis using AUC (AbHa))

For MahGol, the results are as follows:

Frequency	T-stat	P-value
LF	0.193669	0.848319
IF	0.954579	0.346190
HF	0.621588	0.539345

Once again, no meaningful difference is observed for this hypothesis in the MahGol trialKey, and the hypothesis cannot be accepted.

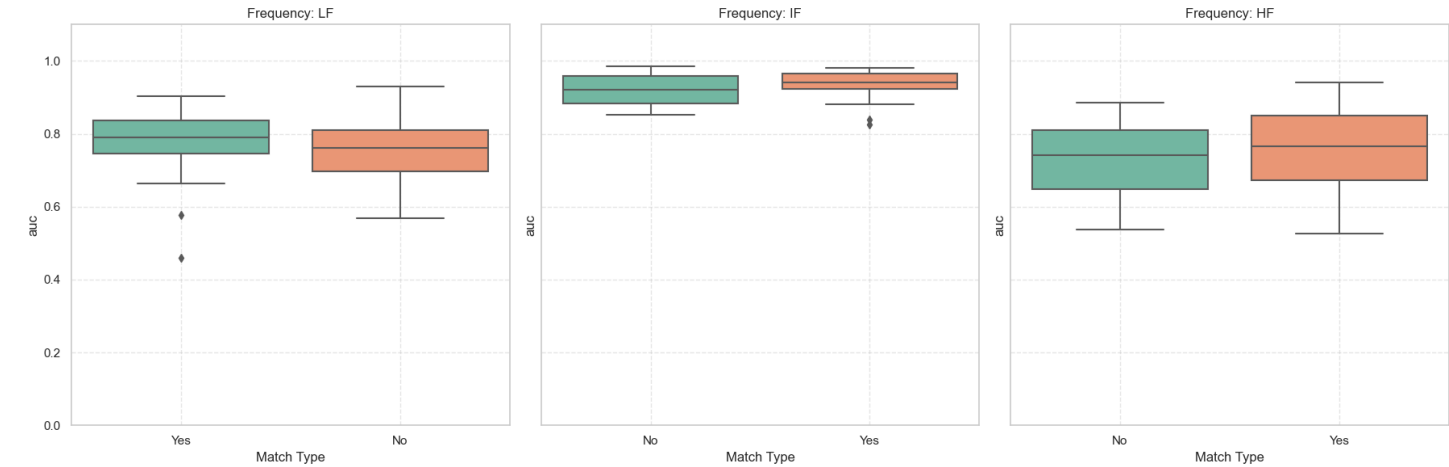


Figure 26 (Boxplot for Analyze Fifth Hypothesis (MahGol))

Make a Hypothesis!

In this section, we formulate additional hypotheses and examine them using Psychometric function and ROC curve.

Sixth hypothesis:

The difference between Conspecific and Heterospecific sensitivity is most pronounced in the LF, HF, IF band.

This hypothesis examines the recognition of one’s own gender at each levelFreq. To test this hypothesis, we use t-test, and the results are as follows:

Sigmoid:

Note that for the sigmoid fitting, some subjects were excluded due to outlier data .

Frequency	T-stat	P-value
LF	0.057176	0.954570
IF	0.140762	0.888549
HF	-1.249991	0.217019

Based on the obtained p-value, we cannot accept the hypothesis. However, in the HF condition, the difference appears to be more noticeable compared to the other frequencies. This observation is also visually supported by the boxplot.

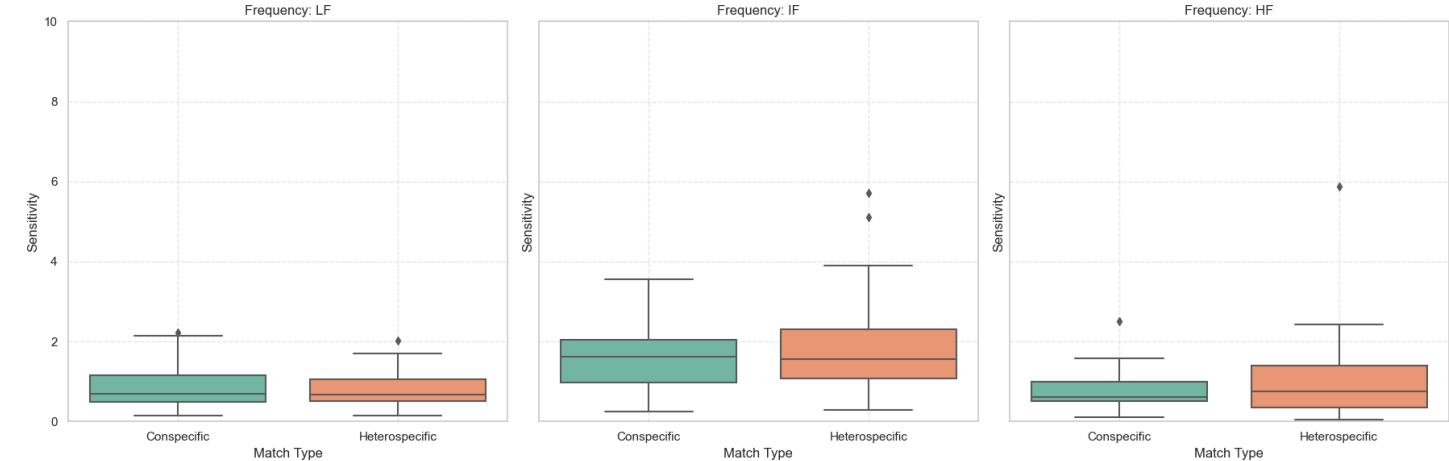


Figure 27 (Boxplot for Analyze Sixth Hypothesis using β)

ROC:

Frequency	T-stat	P-value
LF	-0.351388	0.726317
IF	-1.361438	0.180228
HF	-0.046470	0.963061

Although none of the frequency bands showed a statistically significant difference between conspecific and heterospecific conditions (all p-values > 0.05), the IF band exhibited the lowest p-value (p = 0.18), suggesting a potential trend that may warrant further investigation with larger sample sizes.

The corresponding boxplots are shown below:

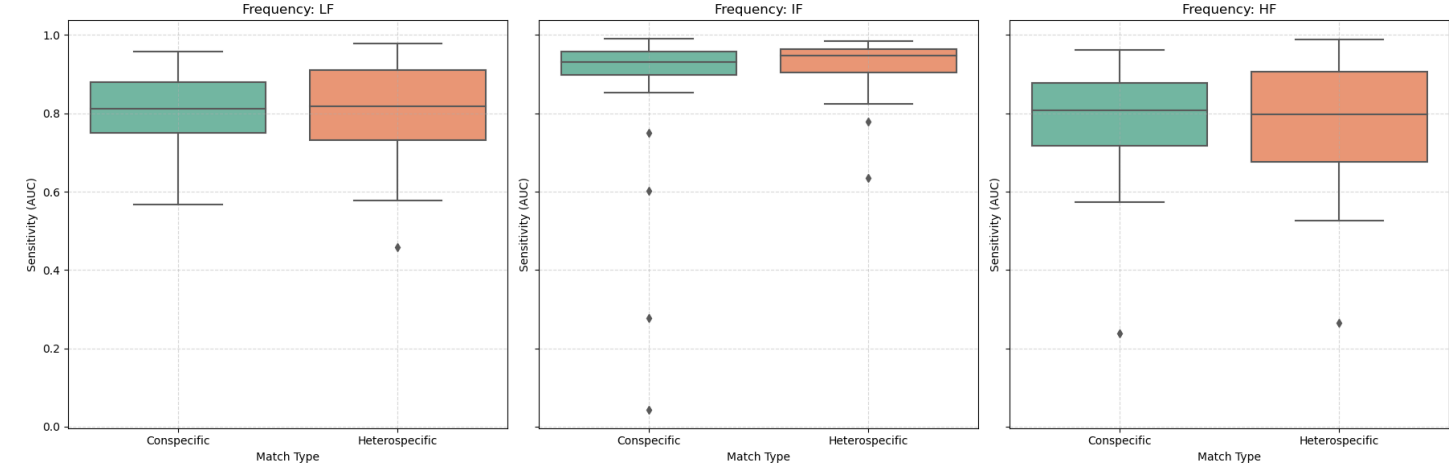


Figure 28 (Boxplot for Analyze Sixth Hypothesis using AUC)

Seventh hypothesis:

The difference between Male and Female sensitivity is most pronounced in the LF, HF, IF band.

In this section, we examine a hypothesis regarding which gender shows differences in sensitivity within each frequency band. This hypothesis is different from the third hypothesis, where we evaluated whether women or men generally perform better in this task. Here, we analyze each frequency band separately.

Sigmoid:

Frequency	T-stat	P-value
LF	0.655483	0.515180
IF	1.313529	0.198735
HF	-1.777583	0.080059

Among the three spatial frequency bands, the HF band shows the strongest trend toward a gender difference in sensitivity (with a p-value close to 0.05). While the result is not statistically significant, it suggests that the male-female difference in identity detection may be most pronounced in the high-frequency band, potentially reaching significance with a larger sample size or in further studies. The corresponding boxplots are shown below:

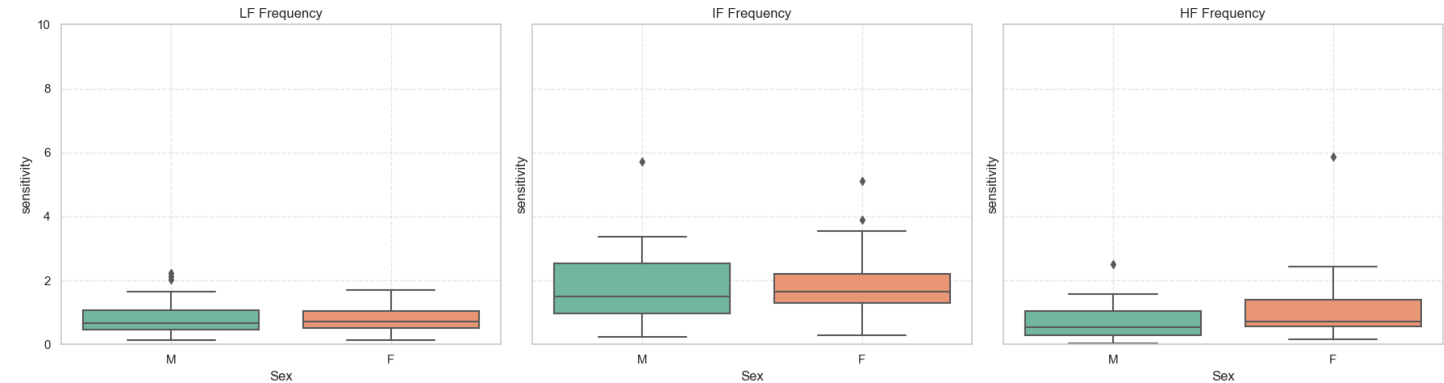


Figure 30 (Boxplot for Analyze Seventh Hypothesis using β)

ROC:

Frequency	T-stat	P-value
LF	-0.847036	0.399884
IF	-1.789211	0.081443
HF	-1.919802	0.059322

The results suggest a trend toward a difference in sensitivity to male versus female stimuli in the HF (high-frequency) band, with p-values approaching significance. In contrast, no substantial differences are observed in the LF and IF bands. These findings, although not statistically significant, echo the pattern seen in the sigmoid fitting results, where the HF band appeared to be the most indicative of identity-related sensitivity differences.

The corresponding boxplots are shown below:

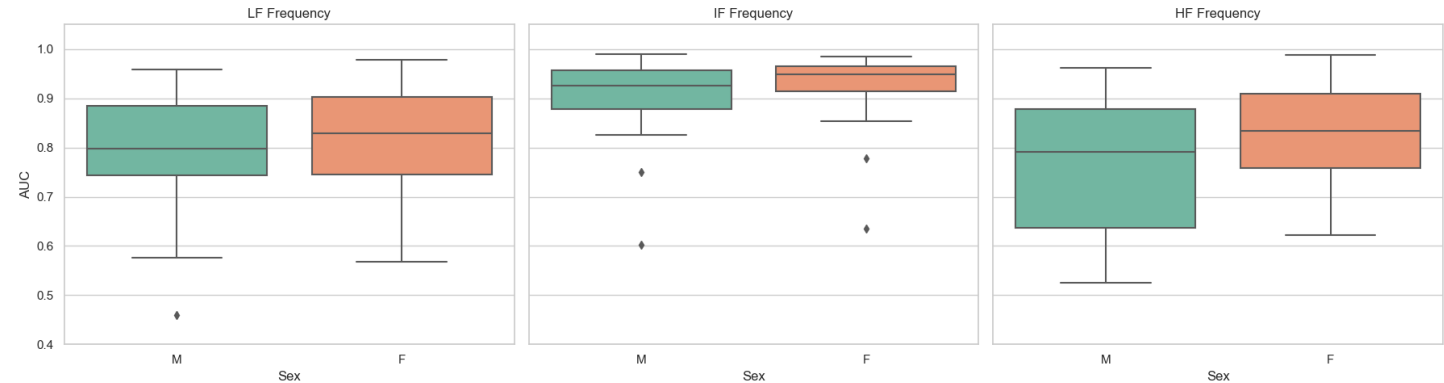


Figure 31 (Boxplot for Analyze Seventh Hypothesis using AUC)

Eighth hypothesis:

Whether age over 25 are significantly better than 25 or less in the task of identity detection.

This hypothesis examines the effect of age on sensitivity and AUC. Age is divided into two categories: subjects over 25 years old and those under 25. Therefore, a t-test is used to evaluate the hypothesis.

Sigmoid:

T-statistic: -0.8713

P-value: 0.3847

Based on the p-value, age does not have a significant effect on sensitivity, and the hypothesis cannot be accepted. The boxplot below also supports the rejection of this hypothesis.

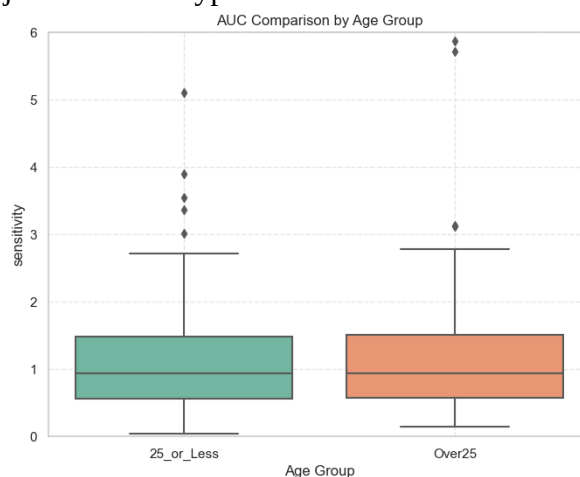


Figure 32 (Boxplot for Analyze Eighth Hypothesis using β)

ROC:

T-statistic: 1.1463

P-value: 0.2532

Similarly, the p-value obtained from the ROC analysis indicates that age has no noticeable impact on AUC, and thus the hypothesis is rejected. The corresponding boxplot is shown below.

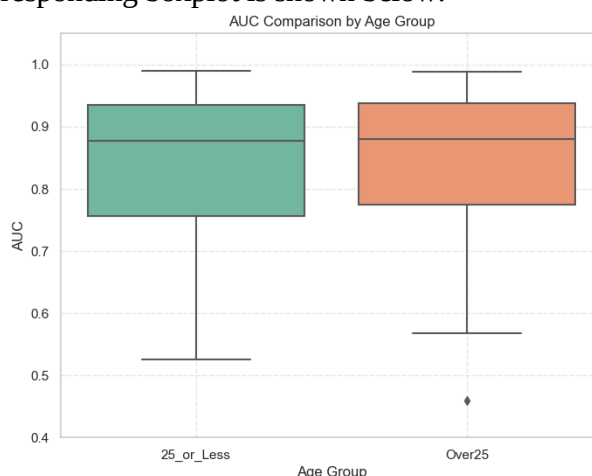


Figure 33 (Boxplot for Analyze Eighth Hypothesis using AUC)

Ninth hypothesis:
The difference in age is most pronounced in the LF, HF, IF band.

In this hypothesis, we aim to investigate the effect of age within each levelFreq. Again, age is divided into two categories, as in the previous hypothesis. This division is based on the average age of the subjects in this task.

Sigmoid:

Frequency	T-stat	P-value
LF	0.104503	0.917241
IF	-1.171584	0.246466
HF	0.702561	0.488501

Although none of the frequency bands showed a statistically significant difference over 25 and less than 25 conditions (all p-values > 0.05), the IF band exhibited the lowest p-value, suggesting a potential trend that may warrant further investigation with larger sample sizes.

The corresponding boxplots are shown below:

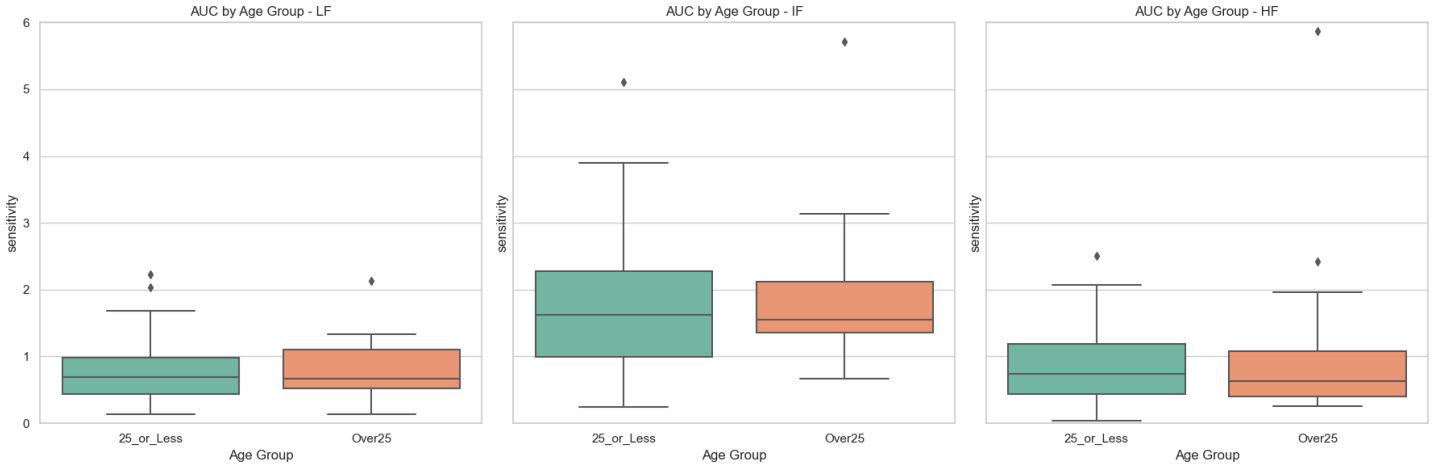


Figure 34 (Boxplot for Analyze Ninth Hypothesis using β)

ROC:

Frequency	T-stat	P-value
LF	0.097159	0.923094
IF	1.413689	0.162301
HF	0.768980	0.444918

Similarly, the p-value obtained from the ROC analysis indicates that age has no noticeable impact on AUC, and thus the hypothesis is rejected. The corresponding boxplot is shown below.

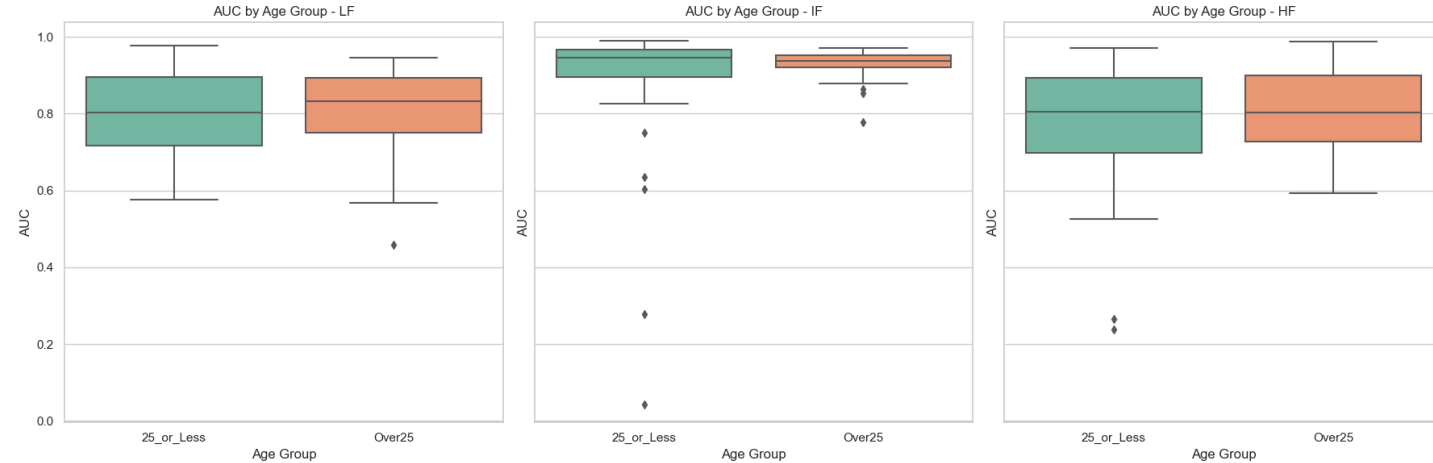


Figure 35 (Boxplot for Analyze Ninth Hypothesis using AUC)

We also conducted another analysis involving age and levelFreq, using a linear regression model with an interaction effect between age and levelFreq. The linear regression model with interaction is a powerful tool for uncovering complex relationships between variables, especially when we suspect that the effect of one independent variable depends on the levels or conditions of another. This model allows us to go beyond simple main effects and better understand interactions between variables. The result of this analysis, based on age categorization into three groups (under 23, between 23 and 27, and over 27), is as follows:

Sigmoid:

OLS Regression Results						
Dep. Variable:	sensitivity	R-squared:	0.073			
Model:	OLS	Adj. R-squared:	0.051			
Method:	Least Squares	F-statistic:	3.289			
Date:	Wed, 16 Apr 2025	Prob (F-statistic):	0.00701			
Time:	01:00:26	Log-likelihood:	-538.88			
No. Observations:	216	AIC:	1090.			
DF Residuals:	210	BIC:	1110.			
DF Model:	5					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	1.3147	2.205	0.596	0.552	-3.033	5.662
levelFreq[T.IF]	0.9054	3.119	0.290	0.772	-5.243	7.054
levelFreq[T.LF]	-0.0416	3.119	-0.013	0.989	-6.190	6.107
age	-0.0165	0.085	-0.193	0.847	-0.185	0.152
age:levelFreq[T.IF]	0.0305	0.121	0.253	0.801	-0.208	0.269
age:levelFreq[T.LF]	-0.0022	0.121	-0.018	0.985	-0.240	0.236

Although the model was statistically significant overall ($p < 0.01$), none of the individual coefficients, including the main effect of age or its interaction with frequencies, were significantly predictive of sensitivity (all $p > 0.7$). Additionally, the non-normality of the residuals increases the likelihood that the assumptions of the regression model have been violated. These results suggest that age and the type of frequency stimulus, either alone or in interaction, are not good predictors of changes in sensitivity.

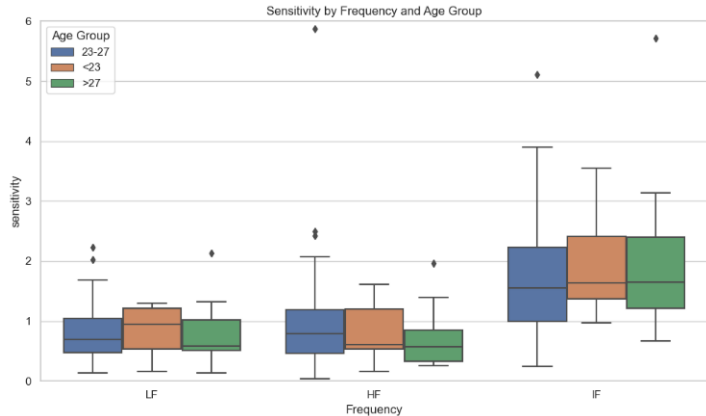


Figure 36 (Boxplot for Analyze Ninth Hypothesis using Regression Model with an Interaction (β))

ROC:

OLS Regression Results						
Dep. Variable:	auc	R-squared:	0.134			
Model:	OLS	Adj. R-squared:	0.115			
Method:	Least Squares	F-statistic:	6.877			
Date:	Wed, 16 Apr 2025	Prob (F-statistic):	5.47e-06			
Time:	03:49:16	Log-likelihood:	135.82			
No. Observations:	228	AIC:	-259.6			
Df Residuals:	222	BIC:	-239.1			
Df Model:	5					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.8949	0.099	9.069	0.000	0.700	1.089
levelFreq[T.IF]	-0.0479	0.140	-0.343	0.732	-0.323	0.227
levelFreq[T.LF]	0.0199	0.140	0.142	0.887	-0.255	0.295
age	-0.0042	0.004	-1.108	0.269	-0.012	0.003
age:levelFreq[T.IF]	0.0064	0.005	1.177	0.240	-0.004	0.017
age:levelFreq[T.LF]	-9.226e-05	0.005	-0.017	0.986	-0.011	0.011

None of the independent variables (age, frequency, or their interaction) had a statistically significant effect on AUC (all p -values > 0.05). Although the overall model was significant ($p < 0.001$), a large portion of the AUC variance remains unexplained. Furthermore, the non-normal distribution of residuals indicates that some model assumptions may have been violated. HF was not excluded from the analysis; rather, it was treated as the reference level in the model. The effects of the other levels (IF and LF) are reported in comparison to HF.

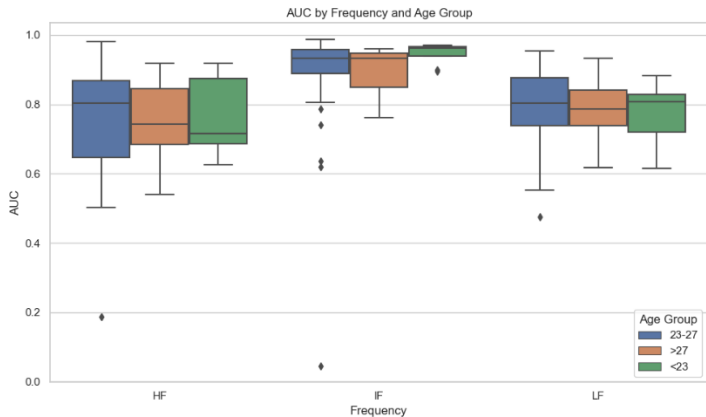


Figure 37 (Boxplot for Analyze Ninth Hypothesis using Regression Model with an Interaction (AUC))

This statistical analysis could have been performed in other sections as well to examine the interaction effects between independent variables and their individual effects.

Tenth hypothesis:
The Effect of Left and Right Hand on Target Selection

In this section, we examine the hypothesis regarding the effect of hand use on image recognition. First, we select the hands used by each subject for target selection in general (not for each trialKey). Here, levelFreq is not important to us. After extracting this information, we perform the test.

Sigmoid:
T-statistic: -1.957367556216599
P-value: 0.06962080415266572
The obtained p-value is close to 0.05, suggesting that with more data, the hypothesis could potentially be supported. The corresponding boxplot is shown below:

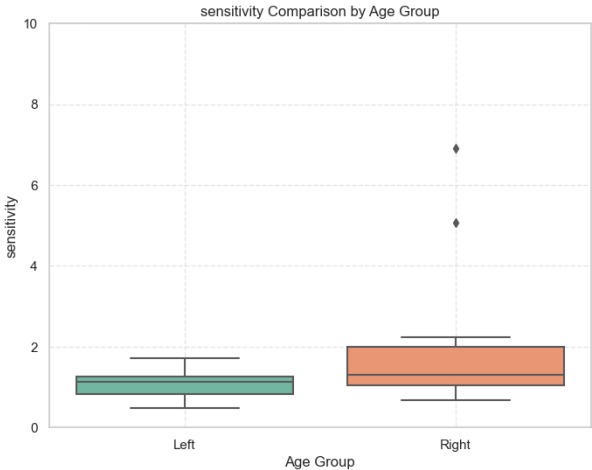


Figure 38 (Boxplot for Analyze Tenth Hypothesis using β)

ROC:
T-statistic: -2.633642600036609
P-value: 0.012371903048146343
Based on the p-value, we can confidently conclude that the hand used does have an impact on selecting the correct face (target), with the right hand having a higher average performance compared to the left hand, suggesting it has a greater effect. However, it is important to note that individuals who are right-handed are not without influence on this result.

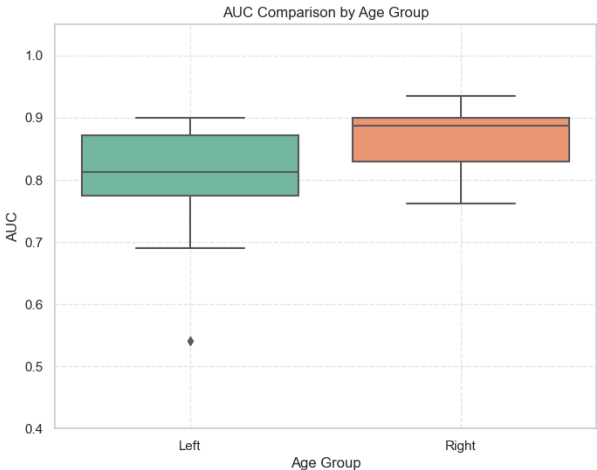


Figure 39 (Boxplot for Analyze Tenth Hypothesis using AUC)

One of the reasons for the difference in p-values is the exclusion of several subjects in the sigmoid fitting analysis, whereas they were not excluded in the ROC analysis. As a result, the amount of data was reduced in the sigmoid fitting, leading to a higher p-value and consequently a different conclusion regarding the rejection of the hypothesis.

Conclusion

In this project, we examined a psychophysics task focused on visual perception. After administering the task to subjects under various conditions and collecting data, we evaluated their perceptual performance using two different methods: the psychometric function and the ROC curve. These approaches provided us with key indices to assess subject performance and test a series of hypotheses. We applied various statistical analyses to determine whether each hypothesis could be accepted or rejected based on the observed data.

These two methods were used as complementary tools, and it is not appropriate to consider one superior to the other. The psychometric function is particularly well-suited for examining individuals' sensitivity to gradual changes in stimuli (such as face morphing), providing insights into both the slope and the decision threshold. In contrast, AUC in the ROC curve measures overall discrimination performance without relying on a specific threshold, making it more robust to noise and imbalanced data. Therefore, combining these two tools offers a more comprehensive understanding of perceptual performance.

Moreover, aside from two hypotheses, the rest showed consistent outcomes (either acceptance or rejection) across both methods. Differences in p-values can be attributed to variations in the datasets used for each test, which affect metrics like the mean and variance. Nonetheless, in both methods, the p-values ended up being either above or below the 0.05 threshold in a consistent manner. Another reason for these differences, as mentioned earlier, is the exclusion of some subjects during sigmoid fitting, which reduced the dataset size and slightly altered the test results.